

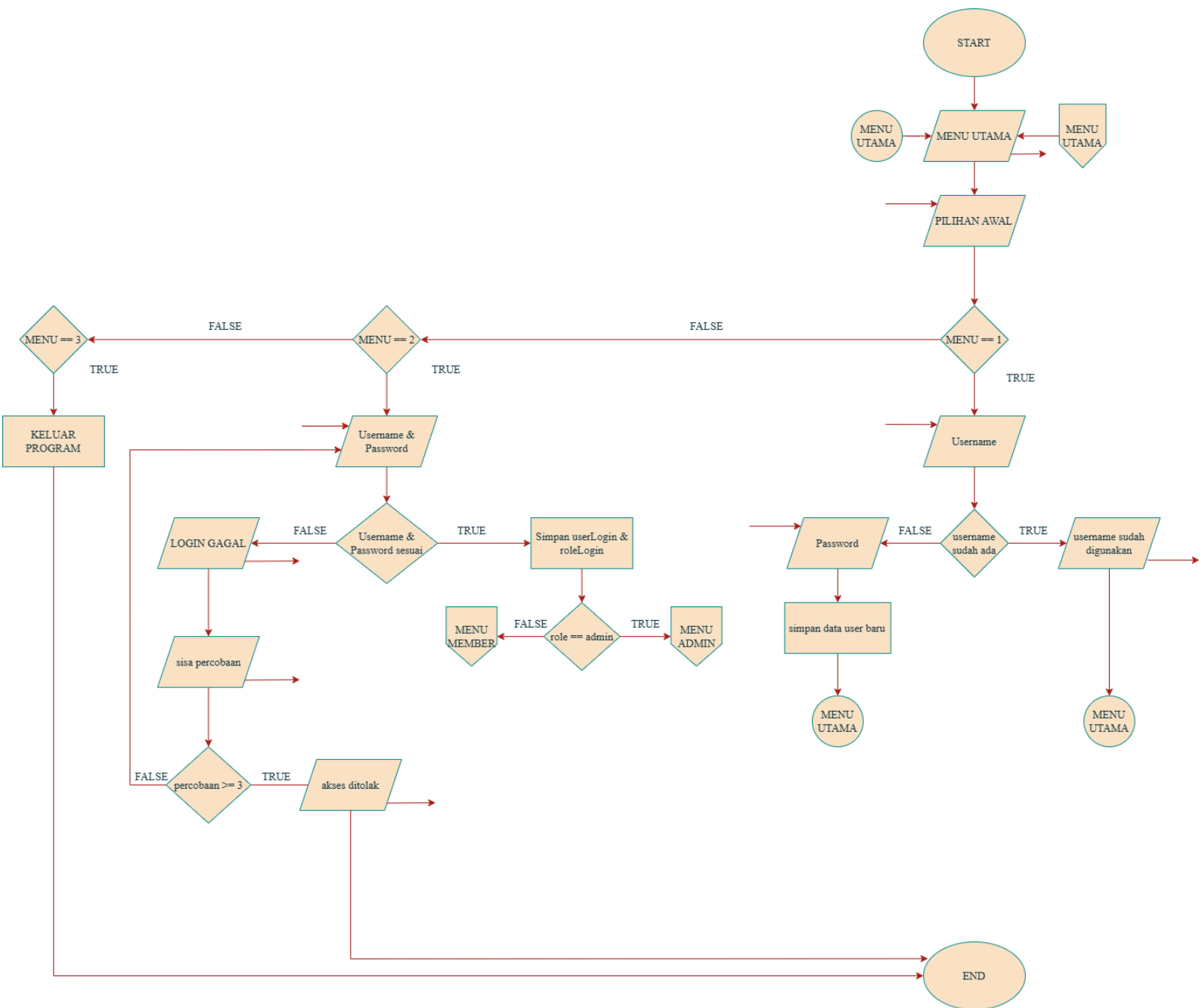
**LAPORAN PRAKTIKUM**  
**POSTTEST 7**  
**ALGORITMA PEMROGRAMAN LANJUT**

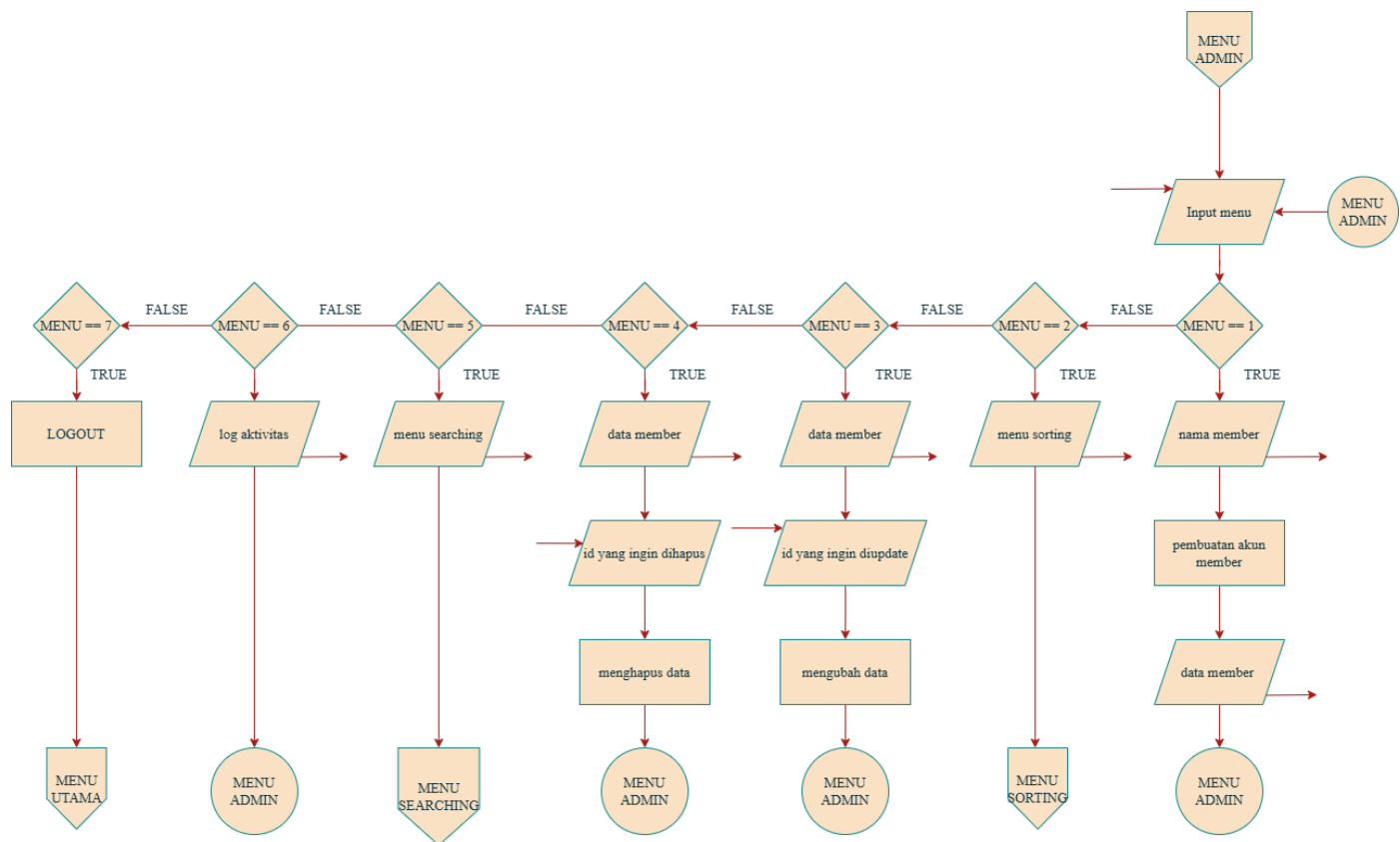


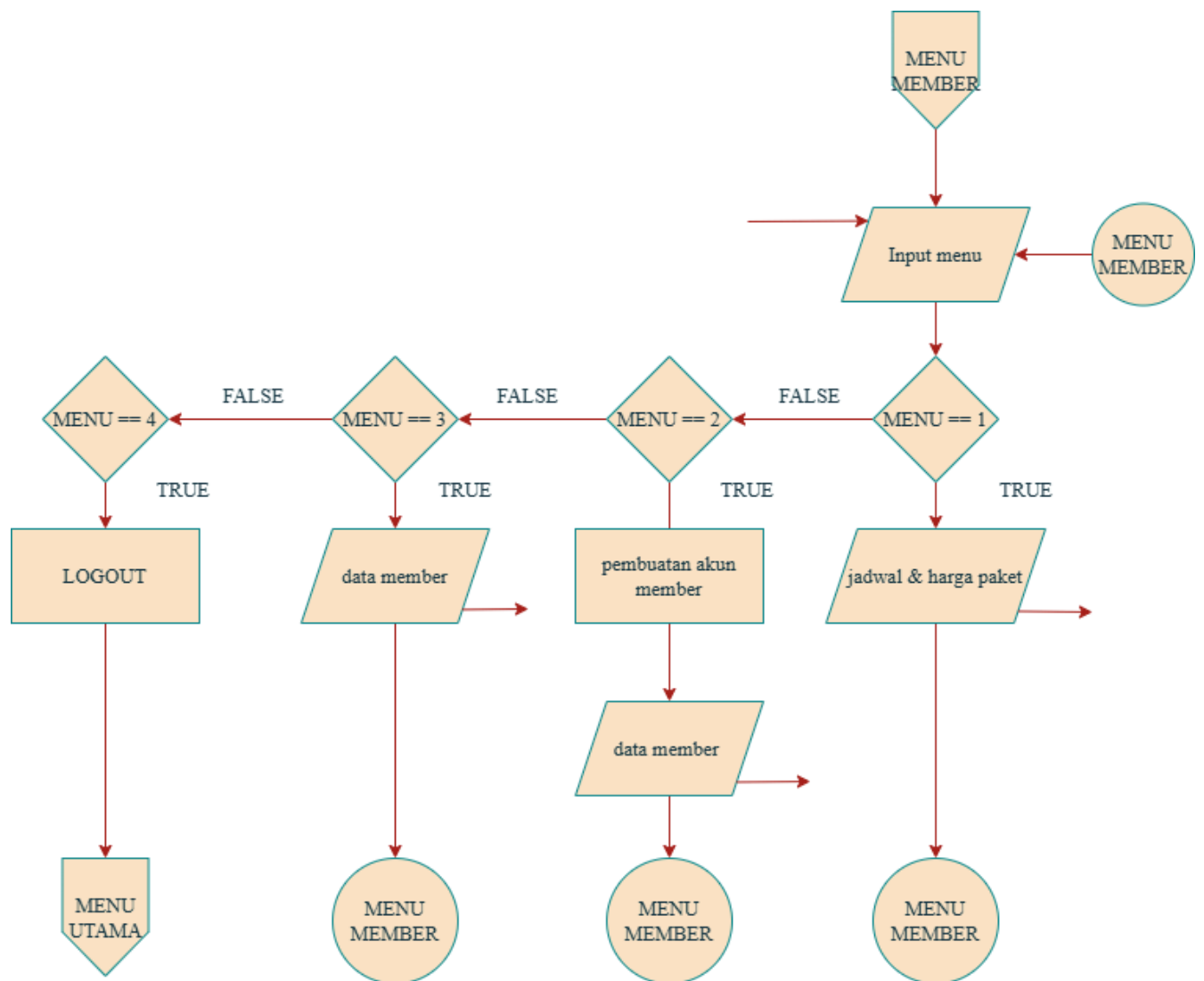
**Disusun oleh:**  
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**Kelas (C1 '25)**

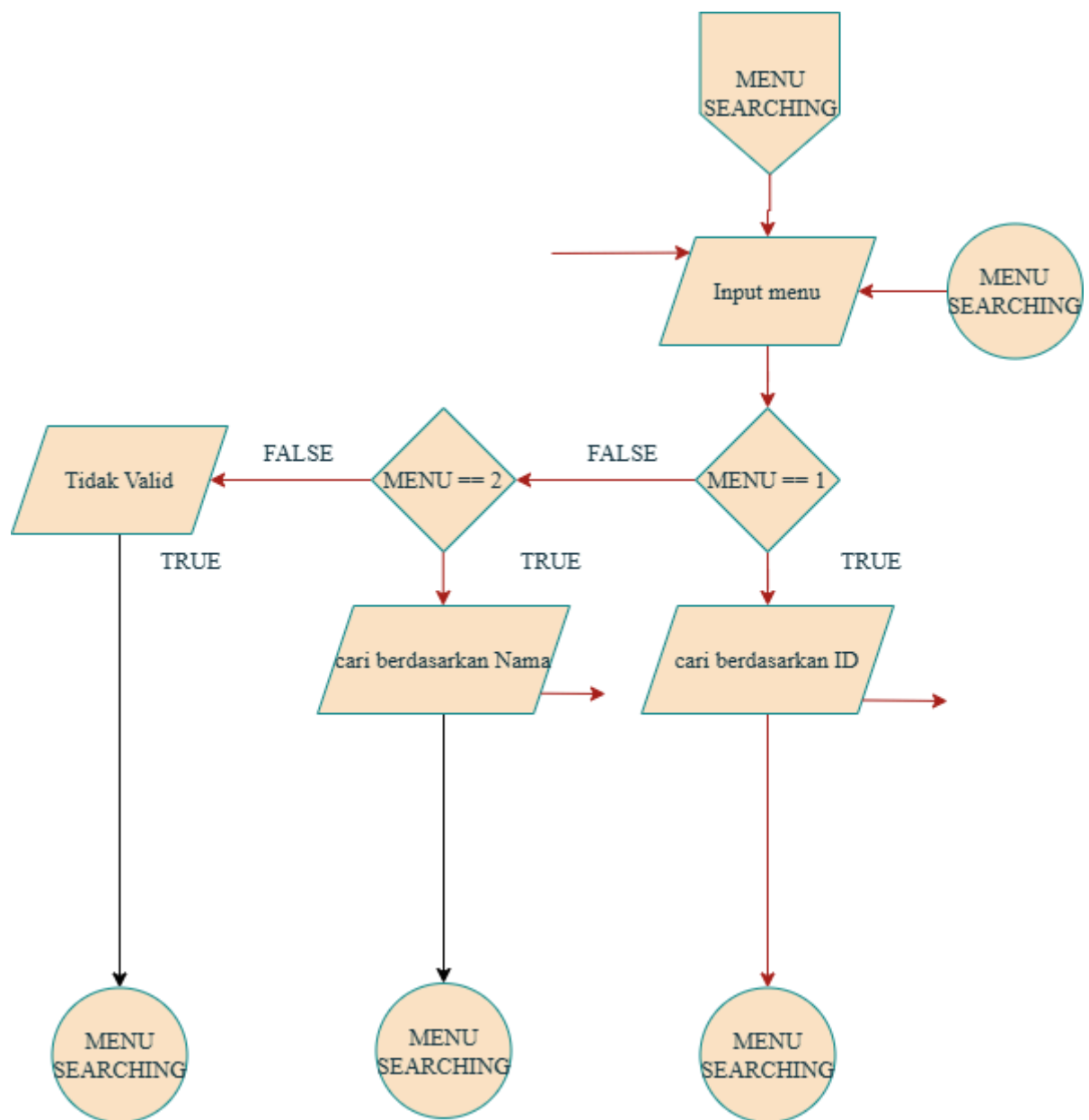
**PROGRAM STUDI INFORMATIKA**  
**UNIVERSITAS MULAWARMAN**  
**SAMARINDA**  
**2026**

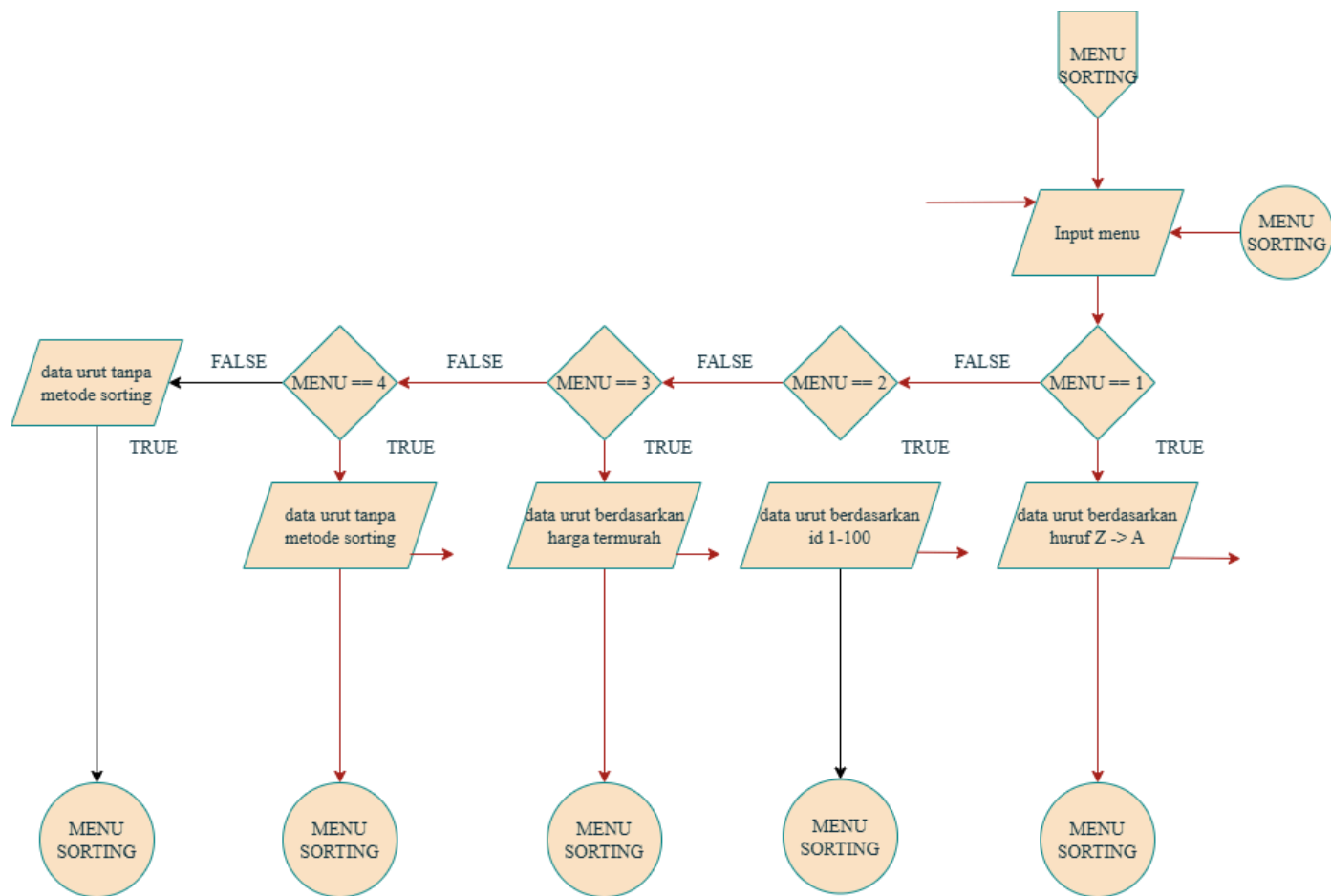
# 1. Flowchart











## 1. Flowchart Menu Utama

- START → Program dimulai
- Tampilkan MENU UTAMA dengan 3 pilihan:
  1. Registrasi Akun Baru
  2. Login
  3. Keluar
- PILIHAN AWAL:
  1. Jika MENU == 1 (REGISTRASI):
    - Input Username
    - Cek apakah username sudah ada?
    - TRUE → Tampil "username sudah digunakan" → Kembali ke MENU UTAMA
    - FALSE → Input Password → Simpan data user baru → Kembali ke MENU UTAMA
  2. B. Jika MENU == 2 (LOGIN):

- Input Username & Password
  - Cek kredensial (username & password sesuai?)
  - FALSE → LOGIN GAGAL → Hitung sisa percobaan
  - Jika percobaan  $\geq 3$  → AKSES DITOLAK → Program END
  - Jika percobaan  $< 3$  → Kembali input login
  - TRUE → Simpan userLogin & roleLogin
  - Cek role == admin?
  - TRUE → Masuk ke MENU ADMIN
  - FALSE → Masuk ke MENU MEMBER
3. Jika MENU == 3 (KELUAR):
- KELUAR PROGRAM → END

## 2. Flowchart Menu Admin

- MENU == 1 → Buat Akun Member Baru
  1. Input nama member
  2. Proses pembuatan akun member
  3. Simpan data member → Kembali ke MENU ADMIN
- MENU == 2 → Tampilkan Seluruh Data Member
  1. Menampilkan semua data dalam array
  2. Tampil data member → Kembali ke MENU ADMIN
- MENU == 3 → Update Data Member
  1. Input id yang ingin diupdate
  2. Tampil data member yang akan diubah
  3. Proses mengubah data (paket, hari, jam, durasi)
  4. Simpan perubahan → Kembali ke MENU ADMIN
- MENU == 4 → Hapus Data Member
  1. Input id yang ingin dihapus
  2. Tampil data member yang akan dihapus
  3. Proses menghapus data (shift array)
  4. Kembali ke MENU ADMIN
- MENU == 5 → LOGOUT → Kembali ke MENU UTAMA

### 3. Flowchart Menu Member

- MENU == 1 → Tampilkan jadwal & harga paket → Kembali ke MENU MEMBER
- MENU == 2 → Pembuatan akun member
  1. Input data member (nama, paket, hari, jam, durasi)
  2. Simpan data member → Kembali ke MENU MEMBER
- MENU == 3 → Tampilkan data member (data diri sendiri) → Kembali ke MENU MEMBER
- MENU == 4 → LOGOUT → Kembali ke MENU UTAMA

### 4. Flowchart Menu Sorting

- MENU == 1 → Dataurut berdasarkan huruf Z -> A (Insertion Sort - Descending) → Kembali ke MENU SORTING
- MENU == 2 → Dataurut berdasarkan id 1-100 (Insertion Sort - Ascending) → Kembali ke MENU SORTING
- MENU == 3 → Dataurut berdasarkan harga termurah (Quick Sort - Ascending) → Kembali ke MENU SORTING
- MENU == 4 → Dataurut tanpa metode sorting (tampilan asli) → Kembali ke MENU SORTING

### 5. Flowchart Menu Searching

- MENU == 1 → Cari data berdasarkan ID (Binary Search) → Kembali ke MENU SEARCHING
- MENU == 2 → Cari data berdasarkan Nama (Linear Search) → Kembali ke MENU SEARCHING
- Tidak Valid → Pilihan menu tidak dikenali → Kembali ke MENU SEARCHING

## 2. Deskripsi Singkat Program

Program ini dibuat dengan tujuan untuk memudahkan pengelolaan data member kelas padel secara terstruktur dan efisien. Program ini memungkinkan admin untuk mengelola data member (create, read, update, delete) serta member untuk mendaftar dan melihat data



diri sendiri, dengan validasi paket sesi padel (SUN/MOON/STAR), pemilihan hari sesuai jadwal, dan perhitungan biaya otomatis berdasarkan durasi member.

### 3. Source Code

#### A. LIBRARY

```
#include <iostream>
#include <iomanip>
#include <stdexcept>
#include <sstream>
#include <vector>
#include <algorithm>
#include <cmath>
using namespace std;

// ===== USER-DEFINED LIBRARY (miloutils.h) =====
#ifndef MILOUTILS_H
#define MILOUTILS_H

namespace MiloUtils {
    void clearScreen() {
        cout << string(50, '\n');
    }

    void printHeader(const string& title) {
        cout << "\n+=====+" << endl;
        cout << "| " << left << setw(37) << title << " |" << endl;
        cout << "+=====+" << endl;
    }

    void printSeparator() {
        cout << "===== " << endl;
    }

    void printSuccess(const string& message) {
        cout << "\n[OK] " << message << endl;
    }

    void printError(const string& message) {
        cout << "\n[ERROR] " << message << endl;
    }

    void printWarning(const string& message) {
        cout << "\n[WARNING] " << message << endl;
    }
}
```

```

void pause() {
    cout << "\nPress Enter to continue...";
    cin.ignore();
}

int getValidatedInput(int minVal, int maxVal, const string& prompt) {
    int input;
    while (true) {
        cout << prompt;
        if (cin >> input) {
            if (input >= minVal && input <= maxVal) {
                return input;
            } else {
                printError("Input harus antara " + to_string(minVal) + "-" + to_string(maxVal));
            }
        } else {
            printError("Input harus berupa angka!");
            cin.clear();
            cin.ignore(10000, '\n');
        }
    }
}

#endif

// ===== ASSERT MACRO FOR DEBUGGING =====
#define ASSERT(kondisi, pesan) \
    if (!(kondisi)) { \
        stringstream ss; \
        ss << "[ASSERT FAILED] " << pesan << " | File: " << __FILE__ \
            << " | Line: " << __LINE__ << " | Function: " << __func__;\
        throw runtime_error(ss.str()); \
    }

// ===== STRUCTS & CONSTANTS =====
struct user {
    string username;
    string password;
    string role;
};

struct infopaketa {
    string hari;
    string jamsesi;
    int bulanMember;
}

```

```

    int harga;
};

struct member {
    int id;
    string nama;
    string username;
    string namapaket;
    infopak detail;
};

#define MAX_MEMBER 100
#define MAX_user 20
#define PERTEMUAN_PER_BULAN 4

// ===== CUSTOM EXCEPTION CLASS =====
class PadelException : public exception {
private:
    string message;
public:
    PadelException(const string& msg) : message(msg) {}
    const char* what() const throw() {
        return message.c_str();
    }
};

// ===== HELPER FUNCTIONS =====
int hitungpertemuan(int bulan) {
    try {
        ASSERT(bulan >= 0, "Bulan tidak boleh negatif");
        if (bulan <= 0) return 0;
        ASSERT(bulan <= 12, "Maksimal durasi 12 bulan");
        return PERTEMUAN_PER_BULAN + hitungpertemuan(bulan - 1);
    } catch (const PadelException& e) {
        throw;
    }
}

string infohari(string namapaket) {
    if (namapaket == "SUN") return "Senin-Jumat";
    else if (namapaket == "MOON") return "Senin-Jumat";
    else if (namapaket == "STAR") return "Sabtu-Minggu";
    else return "-";
}

bool validasihari(string hari, string namapaket) {
    if (namapaket == "SUN" || namapaket == "MOON") {

```

```

        return (hari == "Senin" || hari == "senin" ||
                hari == "Selasa" || hari == "selasa" ||
                hari == "Rabu" || hari == "rabu" ||
                hari == "Kamis" || hari == "kamis" ||
                hari == "Jumat" || hari == "jumat");
    } else if (namapaket == "STAR") {
        return (hari == "Sabtu" || hari == "sabtu" ||
                hari == "Minggu" || hari == "minggu");
    }
    return false;
}

string inputharivalid(string namapaket) {
    string hari;
    while (true) {
        try {
            cout << "MAU PADEL DI HARI APA (" << infohari(namapaket) << "): ";
            getline(cin, hari);

            ASSERT(!hari.empty(), "Hari tidak boleh kosong");

            if (validasihari(hari, namapaket)) {
                if (hari.length() > 0) {
                    hari[0] = toupper(hari[0]);
                    for (int i = 1; i < hari.length(); i++) hari[i] = tolower(hari[i]);
                }
                return hari;
            } else {
                throw PadelException("Hari tidak valid untuk paket ini!");
            }
        } catch (const PadelException& e) {
            cout << "[!] " << e.what() << " ";
            if (namapaket == "SUN" || namapaket == "MOON")
                cout << "Paket weekday hanya bisa: Senin-Jumat" << endl;
            else
                cout << "Paket weekend hanya bisa: Sabtu&Minggu" << endl;
        }
    }
}

int hargapaket(string namapaket, int bulan) {
    try {
        ASSERT(bulan >= 0, "Durasi tidak boleh negatif");
        int hargaPerSesi;
        if (namapaket == "SUN") hargaPerSesi = 450;
        else if (namapaket == "MOON") hargaPerSesi = 500;
        else if (namapaket == "STAR") hargaPerSesi = 650;
    }
}

```

```

        else throw PadelException("Paket tidak valid!");

        int totalPertemuan = hitungpertemuan(bulan);
        return hargaPerSesi * totalPertemuan;
    } catch (const PadelException& e) {
        throw;
    }
}

int hargapaket(string namapaket) {
    if (namapaket == "SUN") return 450;
    else if (namapaket == "MOON") return 500;
    else if (namapaket == "STAR") return 650;
    else return 0;
}

void infopilihanjam(string namapaket) {
    cout << "\n--- PILIH JAM SESI (max 1 pilihan) ---" << endl;
    if (namapaket == "SUN") {
        for (int i = 1; i <= 10; i++)
            cout << i << ". " << (5+i) << ".00-" << (6+i) << ".00" << endl;
    } else if (namapaket == "MOON" || namapaket == "STAR") {
        for (int i = 1; i <= 7; i++)
            cout << i << ". " << (15+i) << ".00-" << (16+i) << ".00" << endl;
    }
}

string jamsesi(string namapaket, int pilihan) {
    try {
        if (namapaket == "SUN" && pilihan >= 1 && pilihan <= 10) {
            int start = 5 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        } else if ((namapaket == "MOON" || namapaket == "STAR") && pilihan >= 1 && pilihan <=
7) {
            int start = 15 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        } else {
            throw PadelException("Pilihan jam tidak valid!");
        }
    } catch (const PadelException& e) {
        cout << "[!] " << e.what() << endl;
        return "-";
    }
    return "-";
}

void tampilkaninfopaket() {

```

```

MiloUtils::printHeader("PAKET SESI DI MILO PADEL");
cout << "1. SUN : weekday, jam 06.00-16.00 | Rp 450.000/sesi" << endl;
cout << "2. MOON : weekday, jam 16.00-23.00 | Rp 500.000/sesi" << endl;
cout << "3. STAR : weekend, jam 16.00-23.00 | Rp 650.000/sesi" << endl;
MiloUtils::printSeparator();
cout << "[FYI] 1 bulan = 4 pertemuan (1x seminggu)" << endl;
MiloUtils::printSeparator();
}

void updateHargaDenganReference(int &harga, int bonus) {
    harga += bonus;
}

void updateNamaDenganPointer(string *namaPtr, string namaBaru) {
    if (namaPtr != nullptr) {
        *namaPtr = namaBaru;
    }
}

void tampilkanDetailMemberDenganPointer(member *mPtr) {
    if (mPtr != nullptr) {
        cout << "\nDetail via Pointer:" << endl;
        cout << " ID: " << mPtr->id << " | Nama: " << mPtr->nama << endl;
        cout << " Paket: " << mPtr->namapaket << " | Harga: Rp " << mPtr->detail.harga << ".000"
<< endl;
    }
}

void demoPointerInternal(member *m, int idx) {
    cout << "\n[Pointer Info] Member #" << idx << ":" << endl;
    cout << " Alamat memori: " << m << endl;
    cout << " Alamat nama: " << &(m->nama) << endl;
    cout << " Alamat harga: " << &(m->detail.harga) << endl;

    int hargaLama = m->detail.harga;
    updateHargaDenganReference(m->detail.harga, 100);
    cout << " Harga setelah update (+100 via &): Rp " << m->detail.harga << ".000" << endl;
    m->detail.harga = hargaLama;

    string namaLama = m->nama;
    updateNamaDenganPointer(&(m->nama), namaLama + " [VIP]");
    cout << " Nama setelah update (via *): " << m->nama << endl;
    m->nama = namaLama;

    tampilkanDetailMemberDenganPointer(m);
}

```

```
// ===== 4. SORTING FUNCTIONS =====

void sortNamaDescending(member arr[], int n) {
    sort(arr, arr + n, [](const member& a, const member& b) {
        return a.nama > b.nama;
    });
}

void sortIdAscending(member arr[], int n) {
    sort(arr, arr + n, [](const member& a, const member& b) {
        return a.id < b.id;
    });
}

void sortHargaAscending(member arr[], int n) {
    sort(arr, arr + n, [](const member& a, const member& b) {
        return a.detail.harga < b.detail.harga;
    });
}

// ===== 5. SEARCHING FUNCTIONS =====

int binarySearchID(member *arr, int n, int target) {
    int low = 0;
    int high = n - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid].id == target) {
            return mid;
        } else if (arr[mid].id < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    return -1;
}

int linearSearchNama(member *arr, int n, string target) {
    for (int i = 0; i < n; i++) {
        if (arr[i].nama == target) {
            return i;
        }
    }
    return -1;
}
```

```

// ===== LOGGING FUNCTIONS =====
vector<string> logAktivitas;

void catatLog(const string& aktivitas) {
    logAktivitas.push_back(aktivitas);
}

void tampilkanLog() {
    if (logAktivitas.empty()) {
        cout << "\n[INFO] Belum ada aktivitas tercatat." << endl;
        return;
    }
    cout << "\n=== LOG AKTIVITAS ===" << endl;
    for (size_t i = 0; i < logAktivitas.size(); i++) {
        cout << "[" << (i+1) << "]" << logAktivitas[i] << endl;
    }
    cout << "=====" << endl;
}

// ===== CRUD FUNCTIONS =====
void buatmember(member members[], int &jumlahmember, string username) {
    try {
        for (int i = 0; i < jumlahmember; i++) {
            if (members[i].username == username) {
                throw PadelException("MEMBER SUDAH ADA! Gunakan menu 'lihat data saya' untuk melihat data sendiri.");
            }
        }
    }

    ASSERT(jumlahmember < MAX_MEMBER, "Kapasitas member penuh");

    cout << "\n--- BUAT AKUN MEMBER ---" << endl;
    member baru;
    baru.id = jumlahmember + 1;
    baru.username = username;

    cout << "Nama: ";
    getline(cin, baru.nama);

    ASSERT(!baru.nama.empty(), "Nama tidak boleh kosong");

    tampilkaninfopaket();
    cout << "Paket (SUN/MOON/STAR): ";
    getline(cin, baru.namapaket);

    if (baru.namapaket != "SUN" && baru.namapaket != "MOON" && baru.namapaket != "STAR") {

```



```

        throw PadelException("PAKET TIDAK VALID! Pilih SUN, MOON, atau STAR.");
    }

    baru.detail.hari = inputharivalid(baru.namapaket);
    infopilihanjam(baru.namapaket);
    cout << "JAM BERAPA (1-" << (baru.namapaket == "SUN" ? "10" : "7") << "): ";
    int pilihanJam;
    cin >> pilihanJam;
    cin.ignore();
    baru.detail.jamsesi = jamsesi(baru.namapaket, pilihanJam);

    cout << "\nMAU JOIN BERAPA BULAN: ";
    cin >> baru.detail.bulanMember;
    cin.ignore();

    ASSERT(baru.detail.bulanMember > 0, "Durasi harus lebih dari 0 bulan");

    baru.detail.harga = hargapaket(baru.namapaket);

    updateHargaDenganReference(baru.detail.harga, 0);

    members[jumlahmember] = baru;
    tampilkanDetailMemberDenganPointer(&members[jumlahmember]);

    jumlahmember++;

    int totalPertemuan = hitungpertemuan(baru.detail.bulanMember);
    int total = hargapaket(baru.namapaket, baru.detail.bulanMember);

    MiloUtils::printSuccess("YEY AKUN MEMBER UDAH ADA!");
    cout << "    Paket      : " << baru.namapaket << endl;
    cout << "    Hari       : " << baru.detail.hari << endl;
    cout << "    Jam sesi   : " << baru.detail.jamsesi << endl;
    cout << "    Durasi     : " << baru.detail.bulanMember << " bulan" << endl;
    cout << "    Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
    cout << "    Total Pertemuan : " << totalPertemuan << "x" << endl;
    cout << "    Harga /Sesi   : Rp " << baru.detail.harga << ".000" << endl;
    cout << "    TOTAL BAYAR   : Rp " << total << ".000 " << endl;

    catatLog("Member baru: " + baru.nama + " (ID: " + to_string(baru.id) + ")");

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
} catch (const exception& e) {
    MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
}
}

```

```

void bacasemuaData(member members[], int jumlahmember) {
    try {
        ASSERT(jumlahmember > 0, "Data member masih kosong");

        cout << "=====" << endl;
        cout << "    SORTING    " << endl;
        cout << "=====" << endl;
        cout << " 1. Nama (Z -> A) [std::sort]    " << endl;
        cout << " 2. ID (1 -> 100) [std::sort]    " << endl;
        cout << " 3. Harga (Termurah) [std::sort]  " << endl;
        cout << " 4. Tanpa Sorting                " << endl;
        cout << "=====" << endl;
        cout << "Pilihan: ";

        int opsi;
        cin >> opsi;
        cin.ignore();

        ASSERT(opsi >= 1 && opsi <= 4, "Pilihan sorting tidak valid");

        member temp[MAX_MEMBER];
        for (int i = 0; i < jumlahmember; i++) temp[i] = members[i];

        if (opsi == 1) {
            sortNamaDescending(temp, jumlahmember);
            cout << "\nData diurutkan: Nama (Z -> A) - std::sort" << endl;
        } else if (opsi == 2) {
            sortIdAscending(temp, jumlahmember);
            cout << "\nData diurutkan: ID (1 -> 100) - std::sort" << endl;
        } else if (opsi == 3) {
            sortHargaAscending(temp, jumlahmember);
            cout << "\nData diurutkan: Harga (Termurah) - std::sort" << endl;
        } else {
            cout << "\nMenampilkan data tanpa sorting" << endl;
        }

        cout <<
        "\n=====
=====
" << endl;

        cout << "                                DAFTAR MEMBER PADEL                                " << endl;
        cout <<
        "=====
=====
" << endl;

        cout << left << setw(4) << "ID"
            << setw(15) << "Nama member"
            << setw(15) << "username"

```

```

        << setw(8) << "Paket"
        << setw(10) << "Hari"
        << setw(12) << "Jam sesi"
        << setw(10) << "Bulan"
        << setw(12) << "Total" << endl;
    cout << "-----" << endl;

    for (int i = 0; i < jumlahmember; i++) {
        int totalPertemuan = hitungpertemuan(temp[i].detail.bulanMember);
        int total = hargapaket(temp[i].namapaket, temp[i].detail.bulanMember);

        cout << left << setw(4) << temp[i].id
            << setw(18) << temp[i].nama
            << setw(15) << temp[i].username
            << setw(8) << temp[i].namapaket
            << setw(12) << temp[i].detail.hari
            << setw(12) << temp[i].detail.jamsesi
            << setw(10) << (to_string(temp[i].detail.bulanMember) + " bln")
            << setw(12) << ("Rp " + to_string(total) + "K") << endl;
    }
    cout <<
    "=====
===== " << endl;

    if (jumlahmember > 0) {
        cout << "\nMenampilkan info member pertama via pointer:" << endl;
        demoPointerInternal(&temp[0], 1);
    }

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
} catch (const exception& e) {
    MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
}
}

void bacadatasaya(member members[], int jumlahmember, string username) {
    try {
        bool ditemukan = false;

        for (int i = 0; i < jumlahmember; i++) {
            if (members[i].username == username) {
                int totalPertemuan = hitungpertemuan(members[i].detail.bulanMember);
                int total = hargapaket(members[i].namapaket, members[i].detail.bulanMember);

                cout << "\n===== " << endl;
                cout << "          DATA MEMBER ANDA          " << endl;
            }
        }
    }
}

```

```

        cout << "===== " << endl;
        cout << "ID          : " << members[i].id << endl;
        cout << "Nama          : " << members[i].nama << endl;
        cout << "Paket          : " << members[i].namapaket << endl;
        cout << "Hari           : " << members[i].detail.hari << endl;
        cout << "Jam sesi       : " << members[i].detail.jamsesi << endl;
        cout << "Durasi         : " << members[i].detail.bulanMember << " bulan" << endl;
        cout << "Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
        cout << "Total Pertemuan : " << totalPertemuan << "x" << endl;
        cout << "Harga /Sesi    : Rp " << members[i].detail.harga << ".000" << endl;
        cout << " TOTAL BAYAR   : Rp " << total << ".000 " << endl;
        cout << "===== " << endl;

        cout << "\n[Pointer Demo] Akses data via pointer:" << endl;
        tampilkanDetailMemberDenganPointer(&members[i]);

        ditemukan = true;
        break;
    }
}

ASSERT(ditemukan, "BELUM JOIN MEMBER! YUK buat akun dulu");

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
} catch (const exception& e) {
    MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
}
}

void updatedatamember(member members[], int jumlahmember) {
    try {
        ASSERT(jumlahmember > 0, "Data member masih kosong");

        cout << "\nDaftar Member:" << endl;
        for (int i = 0; i < jumlahmember; i++) {
            cout << " [" << members[i].id << "] " << members[i].nama
                << " | Paket: " << members[i].namapaket << endl;
        }

        int index;
        cout << "\nID yang mau diupdate: ";
        cin >> index;
        cin.ignore();

        ASSERT(index > 0 && index <= jumlahmember, "ID MEMBER TIDAK VALID");
    }
}

```

```

int idx = index - 1;
cout << "\n--- UPDATE DATA MEMBER ---" << endl;
cout << "Nama: " << members[idx].nama << endl;
cout << "Paket sekarang: " << members[idx].namapaket << endl;

tampilkaninfopaket();
cout << "Paket Baru (SUN/MOON/STAR): ";
getline(cin, members[idx].namapaket);

if (members[idx].namapaket != "SUN" && members[idx].namapaket != "MOON" &&
members[idx].namapaket != "STAR") {
    throw PadelException("PAKET TIDAK VALID!");
}

members[idx].detail.hari = inputharivalid(members[idx].namapaket);
infopilihanjam(members[idx].namapaket);
cout << "Pilih jam sesi (1-" << (members[idx].namapaket == "SUN" ? "10" : "7") << "): ";
int pilihanJam;
cin >> pilihanJam;
cin.ignore();
members[idx].detail.jamsesi = jamsesi(members[idx].namapaket, pilihanJam);

cout << "MAU JOIN BERAPA BULAN: ";
cin >> members[idx].detail.bulanMember;
cin.ignore();

ASSERT(members[idx].detail.bulanMember > 0, "Durasi harus lebih dari 0 bulan");

members[idx].detail.harga = hargapaket(members[idx].namapaket);
updateHargaDenganReference(members[idx].detail.harga, 0);

int totalPertemuan = hitungpertemuan(members[idx].detail.bulanMember);
int total = hargapaket(members[idx].namapaket, members[idx].detail.bulanMember);

MiloUtils::printSuccess("Paket member \'' + members[idx].nama + "\" UPDATE
BERHASIL!");
cout << "    Total Pertemuan : " << totalPertemuan << "x" << endl;
cout << "    TOTAL BAYAR   : Rp " << total << ".000 " << endl;

cout << "\n[Pointer Demo] Member setelah update:" << endl;
demoPointerInternal(&members[idx], members[idx].id);

catatLog("Update member ID " + to_string(members[idx].id));

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
} catch (const exception& e) {

```

```

        MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
    }
}

void deletedatamember(member members[], int &jumlahmember) {
    try {
        ASSERT(jumlahmember > 0, "Data member masih kosong");

        cout << "\nDaftar Member:" << endl;
        for (int i = 0; i < jumlahmember; i++) {
            cout << " [" << members[i].id << "]" << members[i].nama << endl;
        }

        int index;
        cout << "\nID yang mau dihapus: ";
        cin >> index;
        cin.ignore();

        ASSERT(index > 0 && index <= jumlahmember, "ID MEMBER TIDAK VALID");

        cout << "\n[Pointer Demo] Member yang akan dihapus:" << endl;
        demoPointerInternal(&members[index - 1], index);

        for (int i = index - 1; i < jumlahmember - 1; i++) {
            members[i] = members[i + 1];
        }
        jumlahmember--;

        MiloUtils::printSuccess("MEMBER ID " + to_string(index) + " BERHASIL DIHAPUS!");
        catatLog("Hapus member ID " + to_string(index));

    } catch (const PadelException& e) {
        MiloUtils::printError(e.what());
    } catch (const exception& e) {
        MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
    }
}

void searchingMember(member members[], int jumlahmember) {
    try {
        ASSERT(jumlahmember > 0, "Data member masih kosong");

        cout << "\n===== " << endl;
        cout << "      CARI MEMBER      " << endl;
        cout << "===== " << endl;
        cout << " 1. Cari berdasarkan ID" << endl;
        cout << " 2. Cari berdasarkan Nama" << endl;
    }
}

```

```

cout << "===== " << endl;
cout << "Pilihan: ";

int opsi;
cin >> opsi;
cin.ignore();

ASSERT(opsi >= 1 && opsi <= 2, "Pilihan tidak valid");

member temp[MAX_MEMBER];
for (int i = 0; i < jumlahmember; i++) temp[i] = members[i];

if (opsi == 1) {
    sortIdAscending(temp, jumlahmember);

    int cariID;
    cout << "Masukkan ID yang dicari: ";
    cin >> cariID;

    ASSERT(cariID > 0, "ID harus lebih dari 0");

    int hasil = binarySearchID(temp, jumlahmember, cariID);

    if (hasil != -1) {
        MiloUtils::printSuccess("Data ditemukan!");
        cout << " ID      : " << temp[hasil].id << endl;
        cout << " Nama    : " << temp[hasil].nama << endl;
        cout << " Paket   : " << temp[hasil].namapaket << endl;
        cout << " Indeks  : " << hasil << endl;
    } else {
        MiloUtils::printWarning("ID " + to_string(cariID) + " tidak ditemukan.");
    }
} else if (opsi == 2) {
    string cariNama;
    cout << "Masukkan Nama yang dicari: ";
    getline(cin, cariNama);

    ASSERT(!cariNama.empty(), "Nama tidak boleh kosong");

    int hasil = linearSearchNama(temp, jumlahmember, cariNama);

    if (hasil != -1) {
        MiloUtils::printSuccess("Data ditemukan!");
        cout << " ID      : " << temp[hasil].id << endl;
        cout << " Nama    : " << temp[hasil].nama << endl;
        cout << " Paket   : " << temp[hasil].namapaket << endl;
        cout << " Indeks  : " << hasil << endl;
    }
}

```

```

    } else {
        MiloUtils::printWarning("Nama \" + cariNama + "\" tidak ditemukan.");
    }
}

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
} catch (const exception& e) {
    MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
}
}
}

```

## B. MENU ADMIN

```

void menuAdmin(member members[], int &jumlahmember, string userLogin) {
    int pilihan;
    bool menuberjalan = true;

    do {
        try {
            cout << endl;
            MiloUtils::printHeader("ADMIN MENU - " + userLogin);

            cout << "1. Buat Akun Member Baru    " << endl;
            cout << "2. Tampilkan Semua Data      " << endl;
            cout << "3. Update Paket Member    " << endl;
            cout << "4. Hapus Member          " << endl;
            cout << "5. Cari Member           " << endl;
            cout << "6. Lihat Log Aktivitas   " << endl;
            cout << "7. Logout                " << endl;
            MiloUtils::printSeparator();
            cout << "Pilih menu (1-7): ";
            cin >> pilihan;
            cin.ignore();

            ASSERT(pilihan >= 1 && pilihan <= 7, "PILIHAN TIDAK VALID! hanya 1-7");

            switch (pilihan) {
                case 1: buatmember(members, jumlahmember, "admin_managed"); break;
                case 2: bacasemuadata(members, jumlahmember); break;
                case 3: updatedatamember(members, jumlahmember); break;
                case 4: deletedatamember(members, jumlahmember); break;
                case 5: searchingMember(members, jumlahmember); break;
                case 6: tampilkanLog(); MiloUtils::pause(); break;
                case 7:

```



```

        MiloUtils::printSuccess("BERHASIL LOGOUT! kembali ke menu utama.");
        menuberjalan = false;
        break;
    }

    } catch (const PadelException& e) {
        MiloUtils::printError(e.what());
    } catch (const exception& e) {
        MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
    }

} while (menuberjalan);
}

```

### C. MENU MEMBER

```

void menuMember(member members[], int &jumlahmember, string userLogin) {
    int pilihan;
    bool menuberjalan = true;

    do {
        try {
            cout << endl;
            MiloUtils::printHeader("MEMBER MENU - " + userLogin);

            cout << "1. Lihat Jadwal & Harga Paket  " << endl;
            cout << "2. Buat Akun Member          " << endl;
            cout << "3. Lihat Data Saya              " << endl;
            cout << "4. Logout                          " << endl;
            MiloUtils::printSeparator();
            cout << "Pilih menu (1-4): ";
            cin >> pilihan;
            cin.ignore();

            ASSERT(pilihan >= 1 && pilihan <= 4, "PILIHAN TIDAK VALID! hanya 1-4");

            switch (pilihan) {
                case 1: tampilkaninfopakets(); break;
                case 2: buatmember(members, jumlahmember, userLogin); break;
                case 3: bacadasasaya(members, jumlahmember, userLogin); break;
                case 4:
                    MiloUtils::printSuccess("BERHASIL LOGOUT! kembali ke menu utama.");
                    menuberjalan = false;
                    break;
            }
        }
    }
}

```

```

    } catch (const PadelException& e) {
        MiloUtils::printError(e.what());
    } catch (const exception& e) {
        MiloUtils::printError("Terjadi kesalahan: " + string(e.what()));
    }
}

} while (menuberjalan);
}

```

#### D. MENU UTAMA

```

int main() {
    try {
        user users[MAX_user] = {
            {"risma", "102", "member"},
            {"admin", "milopadel", "admin"}
        };
        int jumlahuser = 2;

        member members[MAX_MEMBER];
        int jumlahmember = 0;

        string inputnama, inputpw;
        string userLogin, roleLogin;

        while (true) {
            try {
                MiloUtils::clearScreen();
                MiloUtils::printHeader("WELCOME TO MILO PADEL");

                cout << "\n--- MENU UTAMA---" << endl;
                cout << "1. Registrasi Akun Baru" << endl;
                cout << "2. Login" << endl;
                cout << "3. Keluar" << endl;
                cout << "Pilihan: ";
                int pilihanawal;
                cin >> pilihanawal;
                cin.ignore();

                ASSERT(pilihanawal >= 1 && pilihanawal <= 3, "PILIHAN TIDAK VALID");

                if (pilihanawal == 1) {
                    cout << "\n--- REGISTRASI AKUN ---" << endl;
                    cout << "username: ";
                    getline(cin, inputnama);

```

```

    bool usernameada = false;
    for (int i = 0; i < jumlahuser; i++) {
        if (inputnama == users[i].username) {
            usernameada = true;
            break;
        }
    }

    ASSERT(!usernameada, "username \"\" + inputnama + "\"" SUDAH DIGUNAKAN");

    cout << "Password: ";
    getline(cin, inputpw);

    ASSERT(inputpw.length() >= 3, "Password minimal 3 karakter");

    users[jumlahuser].username = inputnama;
    users[jumlahuser].password = inputpw;
    users[jumlahuser].role = "user";
    jumlahuser++;

    MiloUtils::printSuccess("REGIS BERHASIL! YUK LOGIN");

} else if (pilihanawal == 2) {
    cout << "\n--- LOGIN user ---" << endl;

    bool loginSukses = false;

    for (int percobaan = 0; percobaan < 3; percobaan++) {
        cout << "username: ";
        getline(cin, inputnama);
        cout << "Password: ";
        getline(cin, inputpw);

        bool usnbenar = false;
        bool pwbenar = false;

        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usnbenar = true;
                if (inputpw == users[i].password) {
                    pwbenar = true;
                    userLogin = inputnama;
                    roleLogin = users[i].role;
                    break;
                }
            }
        }
    }
}

```

```

    }

    if (usnbenar && pwbenar) {
        loginSukses = true;
        break;
    }

    MiloUtils::printError("LOGIN GAGAL");
    if (!usnbenar) {
        cout << "    -> USERNAME GA ADA NIH!" << endl;
    } else {
        cout << "    -> PASSWORD SALAH!" << endl;
    }

    ASSERT(percobaan < 2, "AKSES DITOLAK! GAGAL LOGIN 3x");

    cout << "    -> Sisa percobaan: " << (2 - percobaan) << endl << endl;
}

ASSERT(loginSukses, "Login gagal");

if (roleLogin == "admin") {
    menuAdmin(members, jumlahmember, userLogin);
} else {
    menuMember(members, jumlahmember, userLogin);
}

} else if (pilihanawal == 3) {
    MiloUtils::printSuccess("THANK YOU! see you milovers.");
    break;
}

} catch (const PadelException& e) {
    MiloUtils::printError(e.what());
    MiloUtils::pause();
} catch (const exception& e) {
    MiloUtils::printError("Terjadi kesalahan sistem: " + string(e.what()));
    MiloUtils::pause();
}
}

} catch (const exception& e) {
    MiloUtils::printError("Fatal Error: " + string(e.what()));
    return 1;
}

return 0;

```

```
}
```

## 4. Hasil Output

```
+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN user ---
username: dapa
Password: 101

[ERROR] LOGIN GAGAL
      -> PASSWORD SALAH!
      -> Sisa percobaan: 2

username: dapa
Password: 102

[ERROR] LOGIN GAGAL
      -> PASSWORD SALAH!
      -> Sisa percobaan: 1

username: dapa
Password: 103

[ERROR] LOGIN GAGAL
      -> PASSWORD SALAH!

[ERROR] Terjadi kesalahan sistem: [ASSERT FAILED] AKSES DITOLAK! GAGAL LOGIN 3x | File: 2509106102-RISMAYANTY-PT-7.cpp | Line: 874 | Function: main
Press Enter to continue...
```

Gambar 4.1 Hasil Output

```
+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
username: dapa
Password: 100

[OK] REGIS BERHASIL! YUK LOGIN
```

Gambar 4.2 Hasil Output

```
+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN user ---
username: dapa
Password: 100

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): █
```

Gambar 4.3 Hasil Output

```

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): 1

+=====+
| PAKET SESI DI MILO PADEL |
+=====+
1. SUN : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====

```

Gambar 4.4 Hasil Output

```

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: dapa

+=====+
| PAKET SESI DI MILO PADEL |
+=====+
1. SUN : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====
Paket (SUN/MOON/STAR): SUN
MAU PADEL DI HARI APA (Senin-Jumat): Kamis

```

Gambar 4.5 Hasil Output



```

--- PILIH JAM SESI (max 1 pilihan) ---
1. 6.00-7.00
2. 7.00-8.00
3. 8.00-9.00
4. 9.00-10.00
5. 10.00-11.00
6. 11.00-12.00
7. 12.00-13.00
8. 13.00-14.00
9. 14.00-15.00
10. 15.00-16.00
JAM BERAPA (1-10): 2

MAU JOIN BERAPA BULAN: 2

Detail via Pointer:
  ID: 1 | Nama: dapa
  Paket: SUN | Harga: Rp 450.000

[OK] YEY AKUN MEMBER UDAH ADA!
      Paket           : SUN
      Hari            : Kamis
      Jam sesi        : 7.00-8.00
      Durasi          : 2 bulan
      Pertemuan/Bulan : 4x
      Total Pertemuan : 8x
      Harga /Sesi     : Rp 450.000
      TOTAL BAYAR    : Rp 3600.000

```

Gambar 4.6 Hasil Output

```

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): 2

[ERROR] MEMBER SUDAH ADA! Gunakan menu 'lihat data saya' untuk melihat data sendiri.

```

Gambar 4.7 Hasil Output

```

Pilih menu (1-4): 3

=====
                        DATA MEMBER ANDA
=====
ID                : 1
Nama              : dapa
Paket             : SUN
Hari              : Kamis
Jam sesi          : 7.00-8.00
Durasi            : 2 bulan
Pertemuan/Bulan  : 4x
Total Pertemuan   : 8x
Harga /Sesi       : Rp 450.000
TOTAL BAYAR      : Rp 3600.000
=====

[Pointer Demo] Akses data via pointer:

Detail via Pointer:
  ID: 1 | Nama: dapa
  Paket: SUN | Harga: Rp 450.000

```

Gambar 4.8 Hasil Output

```

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): 5

[ERROR] Terjadi kesalahan: [ASSERT FAILED] PILIHAN TIDAK VALID! hanya 1-4 | File: 2509106102-RISMAYANTY-PT-7.cpp | Line: 758 | Function: menuMember

+=====+
| MEMBER MENU - dapa |
+=====+
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=====
Pilih menu (1-4): 4

[OK] BERHASIL LOGOUT! kembali ke menu utama.

```

Gambar 4.9 Hasil Output

```

+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 4

[ERROR] Terjadi kesalahan sistem: [ASSERT FAILED] PILIHAN TIDAK VALID | File: 2509106102-RISMAYANTY-PT-7.cpp | Line: 807 | Function: main
Press Enter to continue...

```

Gambar 4.10 Hasil Output

```

+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 3

[OK] THANK YOU! see you milovers.

```

Gambar 4.11 Hasil Output

```

+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
username: risma

[ERROR] Terjadi kesalahan sistem: [ASSERT FAILED] username "risma" SUDAH DIGUNAKAN | File: 2509106102-RISMAYANTY-PT-7.cpp | Line: 822 | Function: main
Press Enter to continue...

```

Gambar 4.12 Hasil Output

```

+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
2

--- LOGIN user ---
username: risma
Password: 102

```

Gambar 4.13 Hasil Output

```

+=====+
| WELCOME TO MILO PADEL |
+=====+

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN user ---
username: admin
Password: milopadel

+=====+
| ADMIN MENU - admin |
+=====+

1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Cari Member
6. Lihat Log Aktivitas
7. Logout
=====
Pilih menu (1-7): █

```

Gambar 4.14 Hasil Output

```
--- PILIH JAM SESI (max 1 pilihan) ---
1. 16.00-17.00
2. 17.00-18.00
3. 18.00-19.00
4. 19.00-20.00
5. 20.00-21.00
6. 21.00-22.00
7. 22.00-23.00
JAM BERAPA (1-7): 2

MAU JOIN BERAPA BULAN: 2

Detail via Pointer:
  ID: 1 | Nama: mey
  Paket: STAR | Harga: Rp 650.000

[OK] YEY AKUN MEMBER UDAH ADA!
      Paket           : STAR
      Hari            : Sabtu
      Jam sesi        : 17.00-18.00
      Durasi          : 2 bulan
      Pertemuan/Bulan : 4x
      Total Pertemuan : 8x
      Harga /Sesi     : Rp 650.000
      TOTAL BAYAR    : Rp 5200.000
```

Gambar 4.15 Hasil Output

```

+=====+
| ADMIN MENU - admin |
+=====+
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Cari Member
6. Lihat Log Aktivitas
7. Logout
=====
Pilih menu (1-7): 2
=====
          SORTING
=====
1. Nama (Z -> A) [std::sort]
2. ID (1 -> 100) [std::sort]
3. Harga (Termurah) [std::sort]
4. Tanpa Sorting
=====
Pilihan: 1

Data diurutkan: Nama (Z -> A) - std::sort

```

Gambar 4.16 Hasil Output

```

=====
                                DAFTAR MEMBER PADEL
=====
ID  Nama member  username      Paket  Hari    Jam sesi  Bulan  Total
-----
1   mey          admin_managed STAR    Sabtu    17.00-18.00 2 bln   Rp 5200K
=====

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
Alamat memori: 0xc230ff5c30
Alamat nama: 0xc230ff5c38
Alamat harga: 0xc230ff5cdc
Harga setelah update (+100 via &): Rp 750.000
Nama setelah update (via *): mey [VIP]

Detail via Pointer:
ID: 1 | Nama: mey
Paket: STAR | Harga: Rp 650.000

```

Gambar 4.17 Hasil Output

```

    SORTING
=====
1. Nama (Z -> A) [std::sort]
2. ID (1 -> 100) [std::sort]
3. Harga (Termurah) [std::sort]
4. Tanpa Sorting
=====
Pilihan: 2

Data diurutkan: ID (1 -> 100) - std::sort

=====
                                DAFTAR MEMBER PADEL
=====
ID  Nama member      username      Paket  Hari      Jam sesi      Bulan      Total
-----
1   mey              admin_managed  STAR   Sabtu     17.00-18.00  2 bln      Rp 5200K
=====

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
Alamat memori: 0xc230ff5c30
Alamat nama: 0xc230ff5c38
Alamat harga: 0xc230ff5cdc
Harga setelah update (+100 via &): Rp 750.000
Nama setelah update (via *): mey [VIP]

Detail via Pointer:
ID: 1 | Nama: mey
Paket: STAR | Harga: Rp 650.000

```

Gambar 4.18 Hasil Output

```

Pilihan: 3

Data diurutkan: Harga (Termurah) - std::sort

=====
                                DAFTAR MEMBER PADEL
=====
ID  Nama member      username      Paket  Hari      Jam sesi      Bulan      Total
-----
1   mey              admin_managed  STAR   Sabtu     17.00-18.00  2 bln      Rp 5200K
=====

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0xc230ff5c30
  Alamat nama: 0xc230ff5c38
  Alamat harga: 0xc230ff5cdc
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mey [VIP]

Detail via Pointer:
  ID: 1 | Nama: mey
  Paket: STAR | Harga: Rp 650.000

```

Gambar 4.19 Hasil Output

```

Pilihan: 4

Menampilkan data tanpa sorting

=====
                                DAFTAR MEMBER PADEL
=====
ID  Nama member      username      Paket  Hari      Jam sesi      Bulan      Total
-----
1   mey              admin_managed  STAR   Sabtu     17.00-18.00  2 bln      Rp 5200K
=====

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0xc230ff5c30
  Alamat nama: 0xc230ff5c38
  Alamat harga: 0xc230ff5cdc
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mey [VIP]

Detail via Pointer:
  ID: 1 | Nama: mey
  Paket: STAR | Harga: Rp 650.000

```

Gambar 4.20 Hasil Output



```

+=====+
| ADMIN MENU - admin |
+=====+
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Cari Member
6. Lihat Log Aktivitas
7. Logout
=====
Pilih menu (1-7): 3

Daftar Member:
  [1] mey | Paket: STAR

ID yang mau diupdate: 1

--- UPDATE DATA MEMBER ---
Nama: mey
Paket sekarang: STAR

```

Gambar 4.21 Hasil Output

```

[Pointer Demo] Member setelah update:

[Pointer Info] Member #1:
  Alamat memori: 0xc230ffafd0
  Alamat nama: 0xc230ffafd8
  Alamat harga: 0xc230ffb07c
  Harga setelah update (+100 via &): Rp 600.000
  Nama setelah update (via *): mey [VIP]

Detail via Pointer:
  ID: 1 | Nama: mey
  Paket: MOON | Harga: Rp 500.000

```

Gambar 4.22 Hasil Output

```

+=====+
| PAKET SESI DI MILO PADEL |
+=====+
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====
Paket Baru (SUN/MOON/STAR): MOON
MAU PADEL DI HARI APA (Senin-Jumat): rabu

--- PILIH JAM SESI (max 1 pilihan) ---
1. 16.00-17.00
2. 17.00-18.00
3. 18.00-19.00
4. 19.00-20.00
5. 20.00-21.00
6. 21.00-22.00
7. 22.00-23.00
Pilih jam sesi (1-7): 1
MAU JOIN BERAPA BULAN: 1

[OK] Paket member "mey" UPDATE BERHASIL!
      Total Pertemuan : 4x
      TOTAL BAYAR      : Rp 2000.000

```

Gambar 4.23 Hasil Output

```

Pilih menu (1-7): 4

Daftar Member:
  [1] mey

ID yang mau dihapus: 1

[Pointer Demo] Member yang akan dihapus:

[Pointer Info] Member #1:
  Alamat memori: 0xc230ffafd0
  Alamat nama: 0xc230ffafd8
  Alamat harga: 0xc230ffb07c
  Harga setelah update (+100 via &): Rp 600.000
  Nama setelah update (via *): mey [VIP]

Detail via Pointer:
  ID: 1 | Nama: mey
  Paket: MOON | Harga: Rp 500.000

[OK] MEMBER ID 1 BERHASIL DIHAPUS!

```

Gambar 4.24 Hasil Output

```

Pilih menu (1-7): 5

=====
          CARI MEMBER
=====
  1. Cari berdasarkan ID
  2. Cari berdasarkan Nama
=====
Pilihan: 1
Masukkan ID yang dicari: 1

[OK] Data ditemukan!
  ID      : 1
  Nama    : mey
  Paket   : MOON
  Indeks  : 0

```

Gambar 4.25 Hasil Output

```
Pilih menu (1-7): 5

=====
                CARI MEMBER
=====
    1. Cari berdasarkan ID
    2. Cari berdasarkan Nama
=====
Pilihan: 2
Masukkan Nama yang dicari: mwy

[WARNING] Nama "mwy" tidak ditemukan.
```

Gambar 4.26 Hasil Output

```
Pilih menu (1-7): 5

=====
                CARI MEMBER
=====
    1. Cari berdasarkan ID
    2. Cari berdasarkan Nama
=====
Pilihan: 2
Masukkan Nama yang dicari: mey

[OK] Data ditemukan!
ID      : 1
Nama    : mey
Paket   : MOON
Indeks  : 0
```

Gambar 4.27 Hasil Output

```

Pilih menu (1-7): 6

=== LOG AKTIVITAS ===
[1] Member baru: mey (ID: 1)
[2] Update member ID 1
[3] Hapus member ID 1
[4] Member baru: mey (ID: 1)
=====

Press Enter to continue...

```

Gambar 4.28 Hasil Output

```

+=====+
| ADMIN MENU - admin |
+=====+
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Cari Member
6. Lihat Log Aktivitas
7. Logout
=====
Pilih menu (1-7): 8
[ERROR] Terjadi kesalahan: [ASSERT FAILED] PILIHAN TIDAK VALID! hanya 1-7 | File: 2509106102-RISMAYANTY-PT-7.cpp | Line: 716 | Function: menuAdmin

```

Gambar 4.29 Hasil Output

```

+=====+
| ADMIN MENU - admin |
+=====+
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Cari Member
6. Lihat Log Aktivitas
7. Logout
=====
Pilih menu (1-7): 7

[OK] BERHASIL LOGOUT! kembali ke menu utama.

```

Gambar 4.30 Hasil Output

## 5. Langkah-langkah GIT

### 5.1 GIT Add

Git Add adalah perintah yang digunakan untuk menambahkan semua file yang ada di folder tersebut ke staging area. Titik (.) berarti semua file yang berubah atau baru dibuat akan disiapkan untuk disimpan.

```
PS C:\Users\lenov\Downloads\praktikum_apl> git add .
```

Gambar 4.20 Git Add

### 5.2 GIT Commit

Git Commit adalah menyimpan perubahan dari staging area ke local repository. -m "pt7" adalah pesan commit yang menjelaskan perubahan ini untuk "pt7" (pertemuan 7).

- 23 files changed → Ada 3 file yang diubah/ditambahkan
- 908 insertions(+) → Sekitar 908 baris kode baru ditambahkan
- 0 deletion(-) → Tidak ada baris kode yang dihapus

Detail perubahan:

1. create mode 100644  
post-test/post-test-apl-7/2509106102-RISMAYANTY-PT-7.cpp
2. → File post-test PT-7.cpp baru dibuat di folder post-test/post-test-apl-7

```
PS C:\Users\lenov\Downloads\praktikum_apl> git commit -m "pt 7"
[main 1f83445] pt 7
3 files changed, 908 insertions(+)
create mode 100644 post-test/post-test-apl-7/2509106102-RISMAYANTY-PT-7.cpp
create mode 100644 post-test/post-test-apl-7/2509106102-RISMAYANTY-PT-7.exe
```

Gambar 4.21 Git Commit

### 5.3 GIT Push

Git Push adalah mengunggah commit dari local repository ke remote repository (GitHub).

- Enumerating objects: 12, done → Git menghitung 12 objek yang akan diproses
- Counting objects: 100% (12/12), done → Semua 12 objek berhasil dihitung
- Delta compression using up to 16 threads → Git menggunakan kompresi delta dengan 16 thread untuk efisiensi
- Compressing objects: 100% (8/8), done → 8 objek berhasil dikompresi
- Writing objects: 100% (8/8), 738.36 KiB | 5.51 MiB/s, done → 8 objek (ukuran 738.36 KiB) berhasil dikirim dengan kecepatan 5.51 MiB/s
- Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0) → Total 8 objek dengan 3 delta (perbedaan), tidak ada objek yang digunakan ulang
- remote: Resolving deltas: 100% (3/3), completed with 3 local objects → Server GitHub berhasil memproses 3 delta dengan 3 objek lokal

- remote: This repository moved. Please use the new location: → Repository dipindahkan ke lokasi baru
- remote: <https://github.com/rismayanty/praktikum-apl.git> → URL baru repository
- To <https://github.com/rismayanty/praktikum-apl.git> → Berhasil push ke repository GitHub
- 169b00a..1f83445 main -> main → Berhasil push ke repository GitHub dengan commit hash dari 169b00a ke 1f83445 di branch main

```
PS C:\Users\lenov\Downloads\praktikum_apl> git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 738.36 KiB | 5.51 MiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote: This repository moved. Please use the new location:
remote:  https://github.com/rismayanty/praktikum-apl.git
To https://github.com/rismayanty/praktikum.apl.git
169b00a..1f83445  main -> main
```

Gambar 4.22 Git Push